

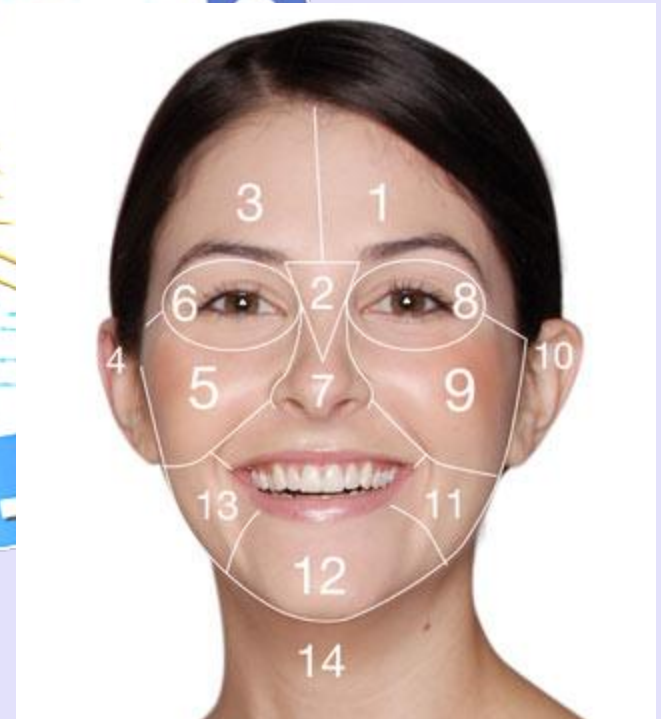
Muscles & nerve supply of face
(Cranial Nerves 5th & 7th)



FACE

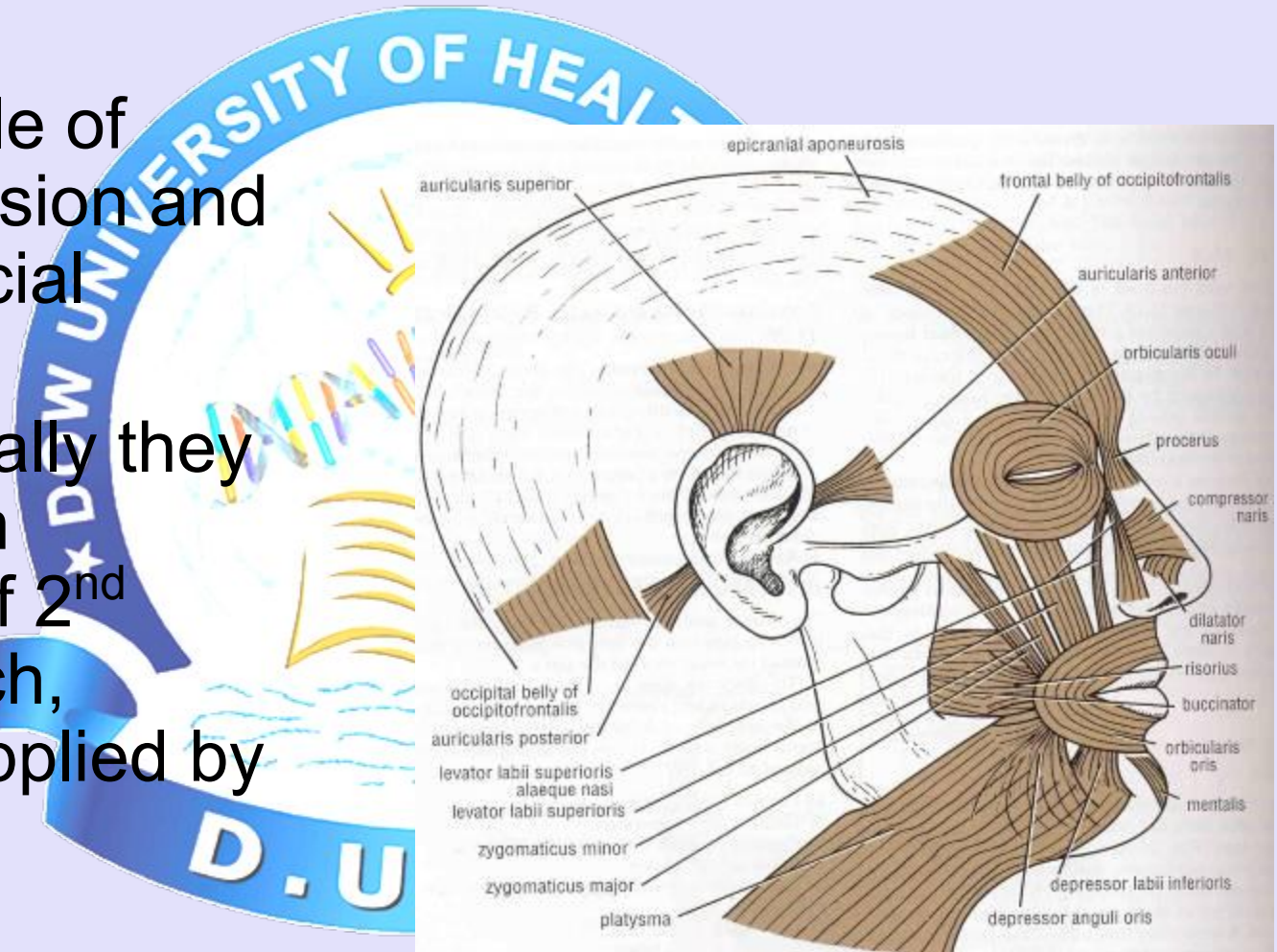
Boundaries:

- Extends superiorly to the hair line, inferiorly to the chin and base of mandible, and on each side to auricle.
- Forehead is common to both scalp and face.
- Skin is very vascular.
- Due to rich vascularity face blush and blanch.
- Wounds of face bleed profusely but heal rapidly.
- Facial skin is rich in sebaceous gland and sweat gland.
- Sebaceous gland keep the skin oily but also cause acne in adult.
- Sweat gland regulate body temperature.



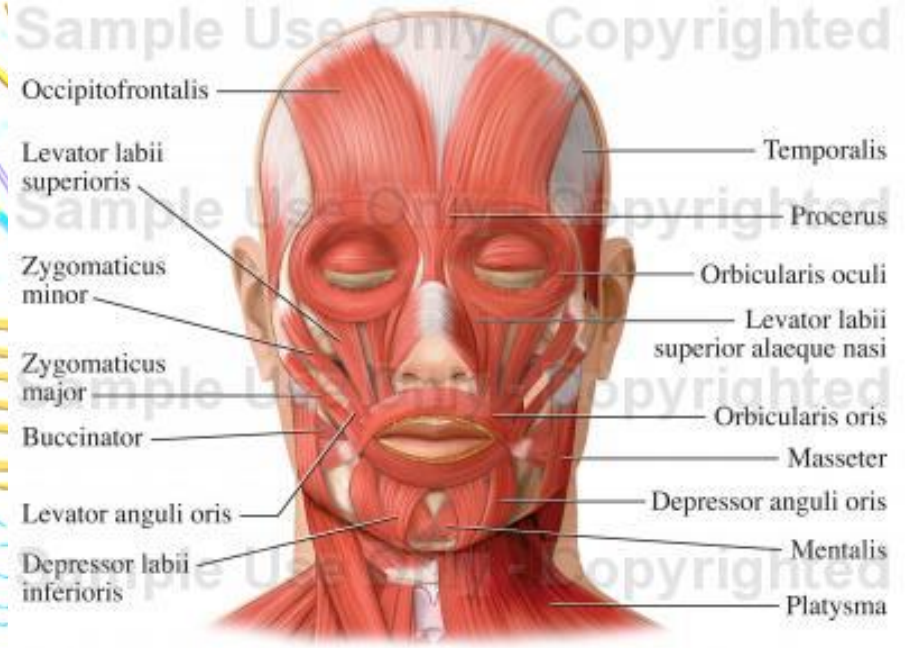
Facial muscles

- Called muscle of facial expression and lie in superficial fascia.
- Embryologically they develop from mesoderm of 2nd branchial arch, therefore supplied by facial nerve.

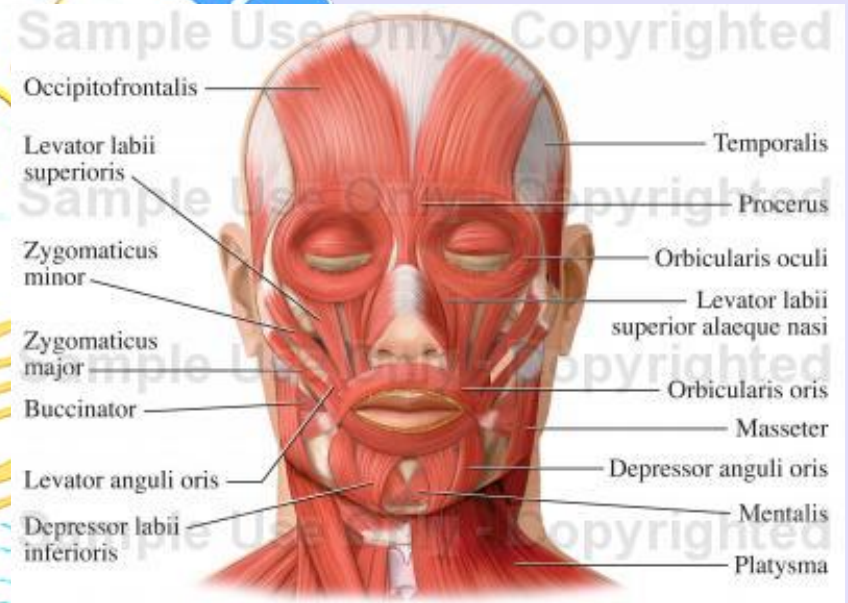


INTRODUCTION

- Muscles of the head can be divided into craniofacial and masticatory groups.
- Craniofacial muscles are often referred to, not very accurately, as 'muscles of facial expression', and are related mainly to the orbital margins and eyelids, the external nose and nostrils, the lips, cheeks and mouth, the pinna, scalp and cervical skin.

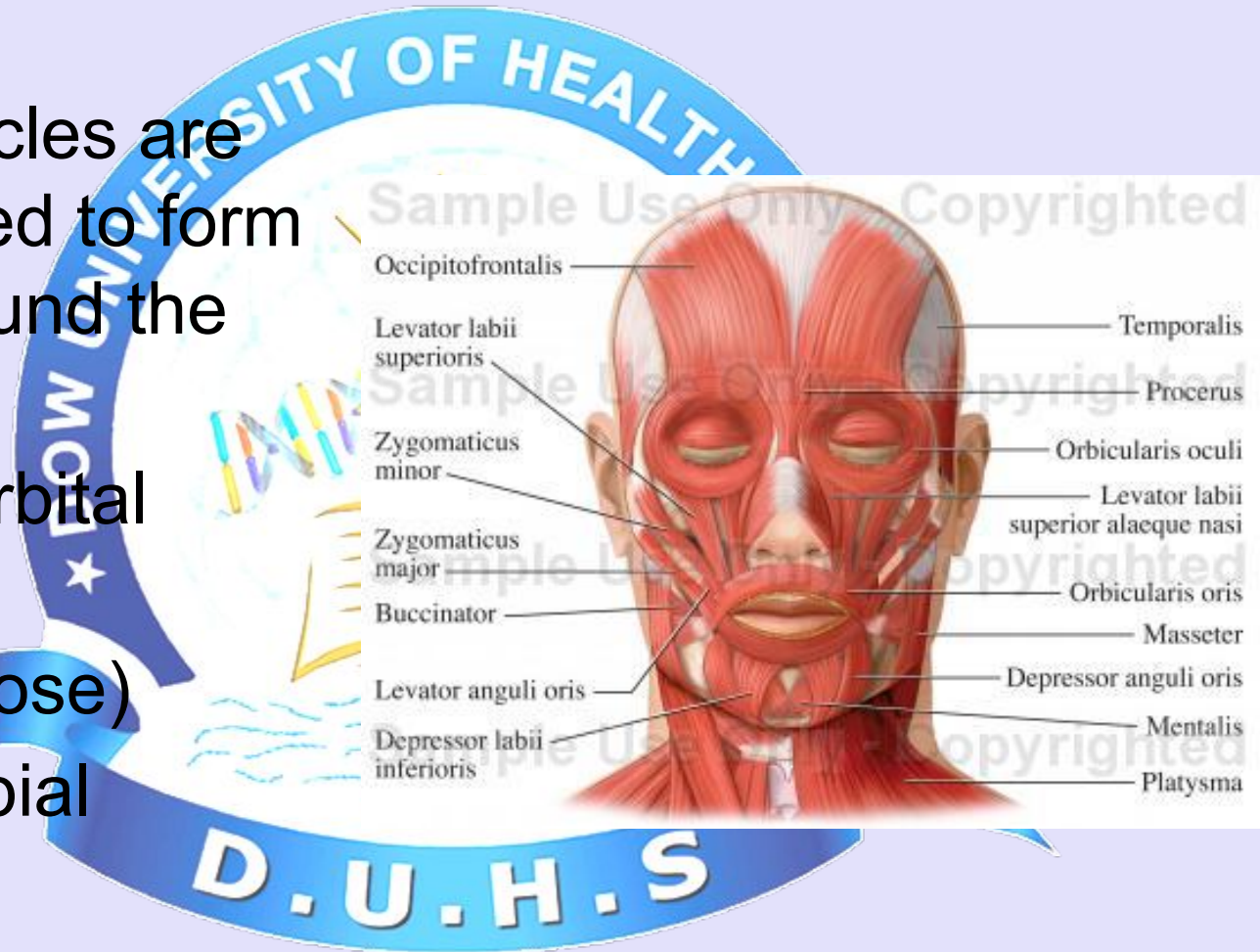


- The organization of the muscles of facial expression differs from that of muscles in most other regions of the body because there is no deep membranous fascia beneath the skin.
- Many small slips of muscle attached to the facial skeleton insert directly into the skin.



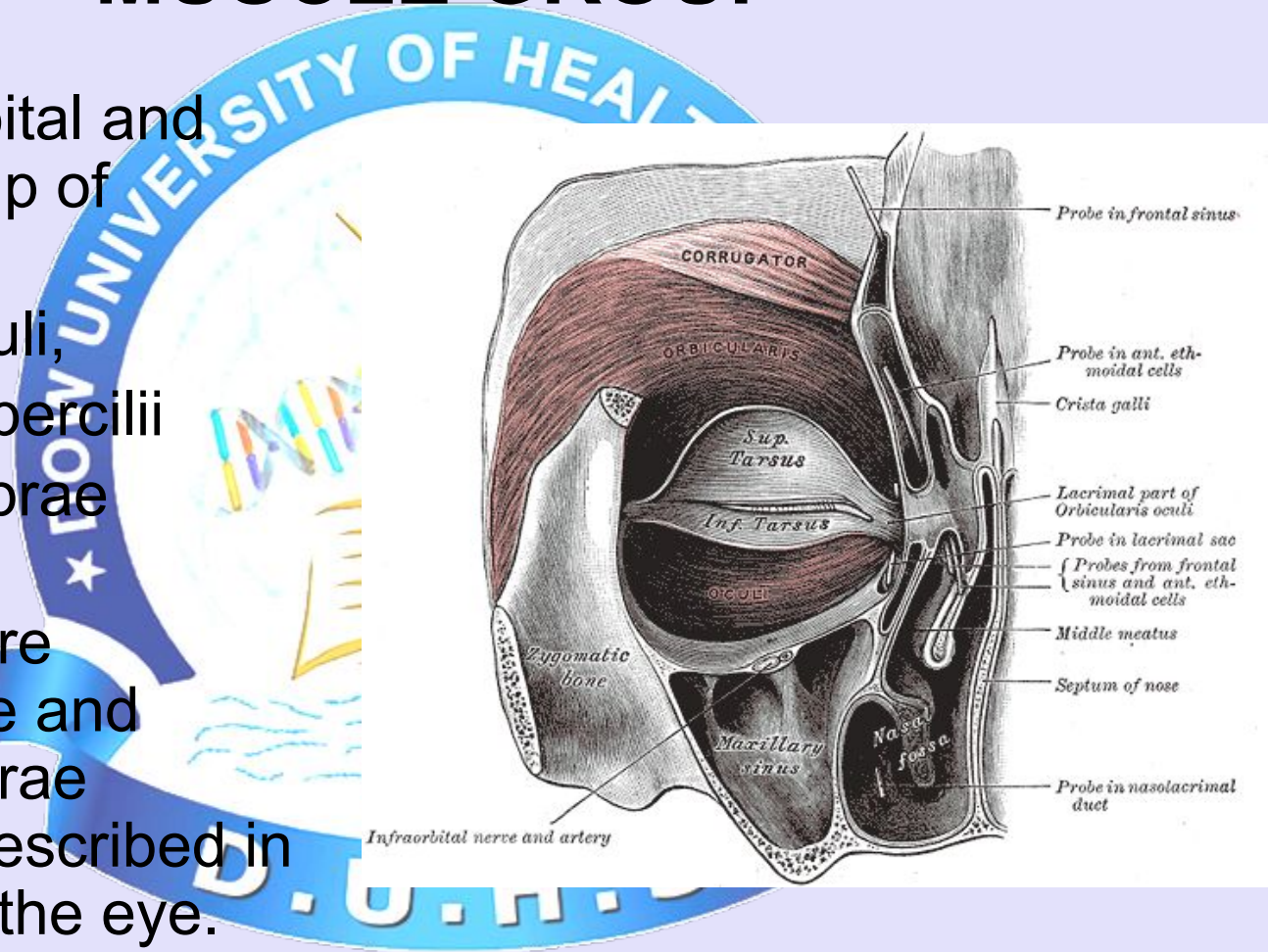
Groups of muscles

- Facial muscles are differentiated to form groups around the orifices
 - Circumorbital (eyes)
 - Nasal (nose)
 - Buccolabial (mouth)



CIRCUMORBITAL AND PALPEBRAL MUSCLE GROUP

- The circumorbital and palpebral group of muscles are
- Orbicularis oculi,
- Corrugator supercilii
- Levator palpebrae superioris.
- The first two are described here and levator palpebrae superioris is described in the context of the eye.



Orbicularis oculi

- 3 parts-
- **Orbital part:**
 - Originate from medial part of medial palpebral ligament and form concentric rings, return to point of origin.

Action –closes the lids tightly.

- **Activated under voluntary control.**

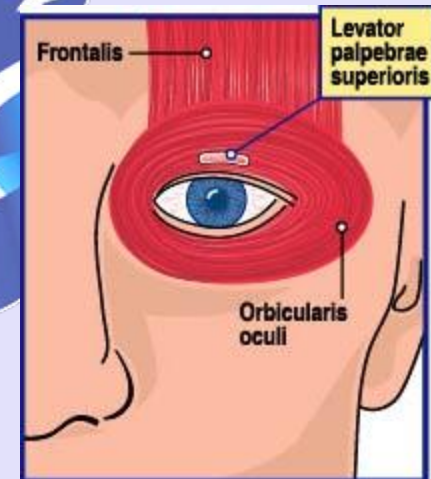
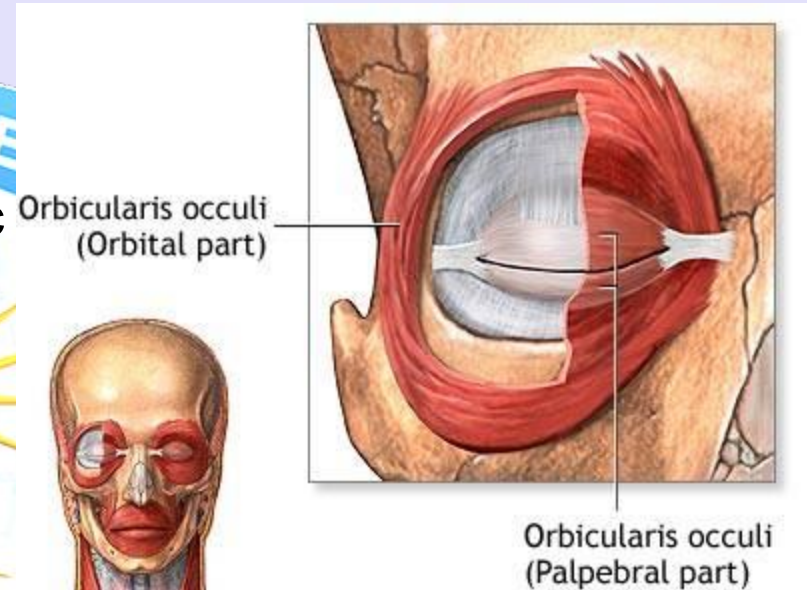
- **Palpebral part:**
 - Originate from lateral part of medial palpebral ligament.
 - Insert into lateral palpebral raphe.

Action-closes the lids gently.

- **Contracted voluntarily and reflexly as in blinking**

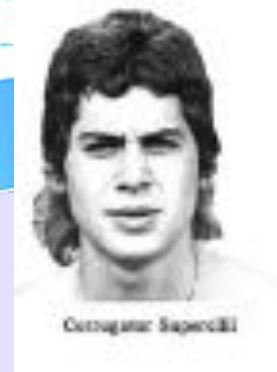
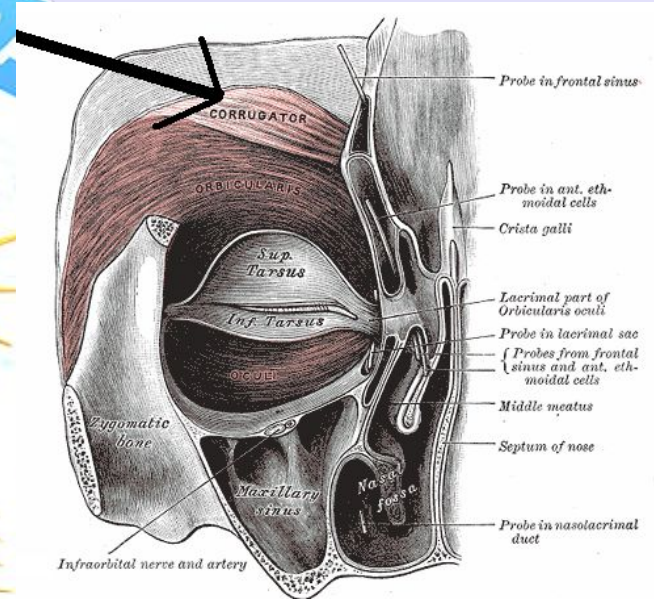
- **Lacrimal part:**
 - Originate from lacrimal fascia & lacrimal bone.
 - Insert into upper & lower tarsi.

Action-dilate lacrimal sac.



Corrugator supercilii

- Corrugator supercilii is a small pyramidal muscle located at the medial end of each eyebrow. .
- **Origin**
 - Medial end of the superciliary arch; its fibers pass upward and lateralward, between the palpebral and orbital portions of the Orbicularis oculi
- **Insertion**
 - Deep surface of the skin, above the middle of the orbital arch .
- **Actions**
 - Draws the eyebrows medially and downwards to shield the eyes in bright sunlight.
 - Involved in frowning

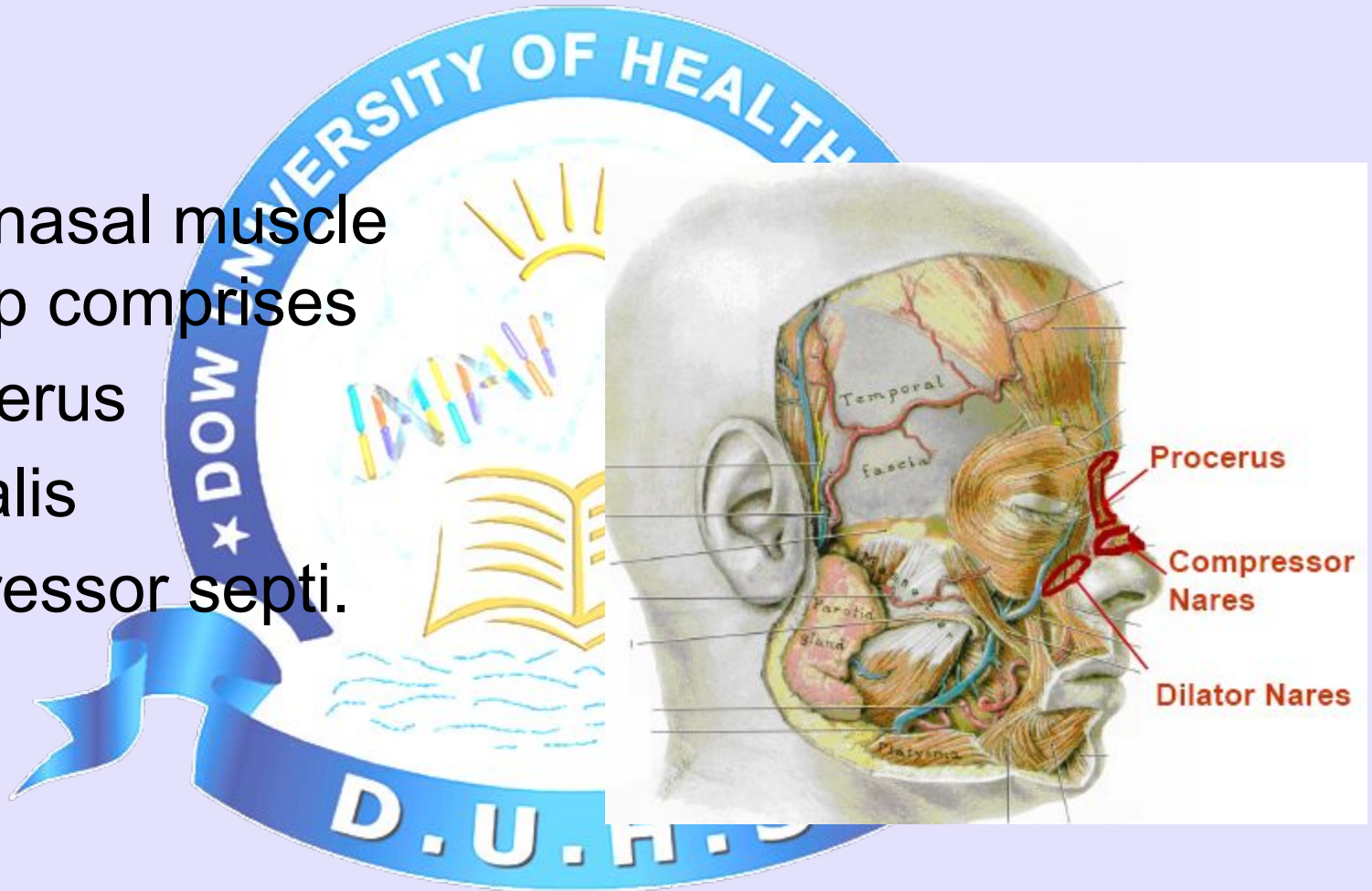


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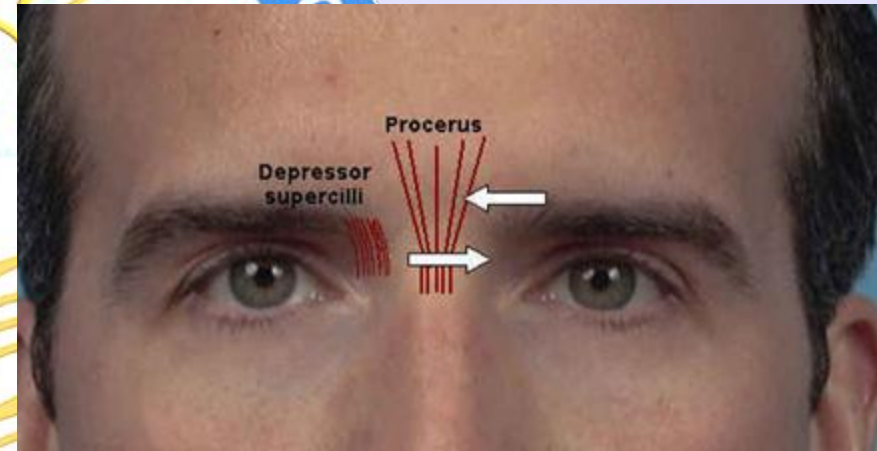
NASAL MUSCLE GROUP

- The nasal muscle group comprises
- Procerus
- Nasalis
- Depressor septi.



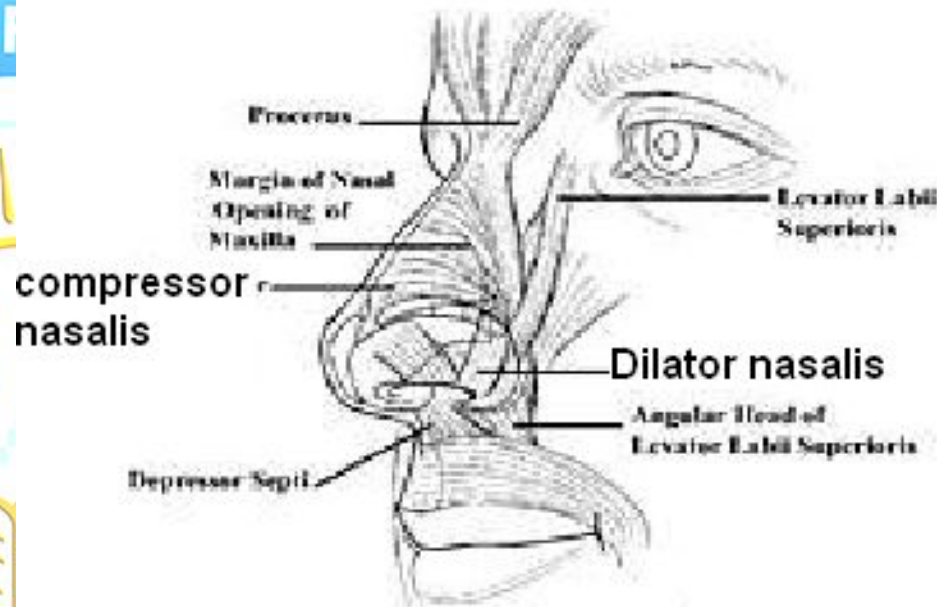
Procerus

- **Procerus** is a small pyramidal slip of muscle deep to the superior orbital nerve, artery and vein.
- **Origin and insertion :**
- It arises by tendinous fibers from the fascia covering the lower part of the nasal bone and upper part of the lateral nasal cartilage.
- It is inserted into the skin over the lower part of the forehead the two eyebrows on either side of the midline, its fibers decussating with those of the Frontalis.
- **Actions**
 - produces transverse wrinkles over the bridge of the nose. It is active in frowning and 'concentration', and helps to reduce the glare of bright sunlight.



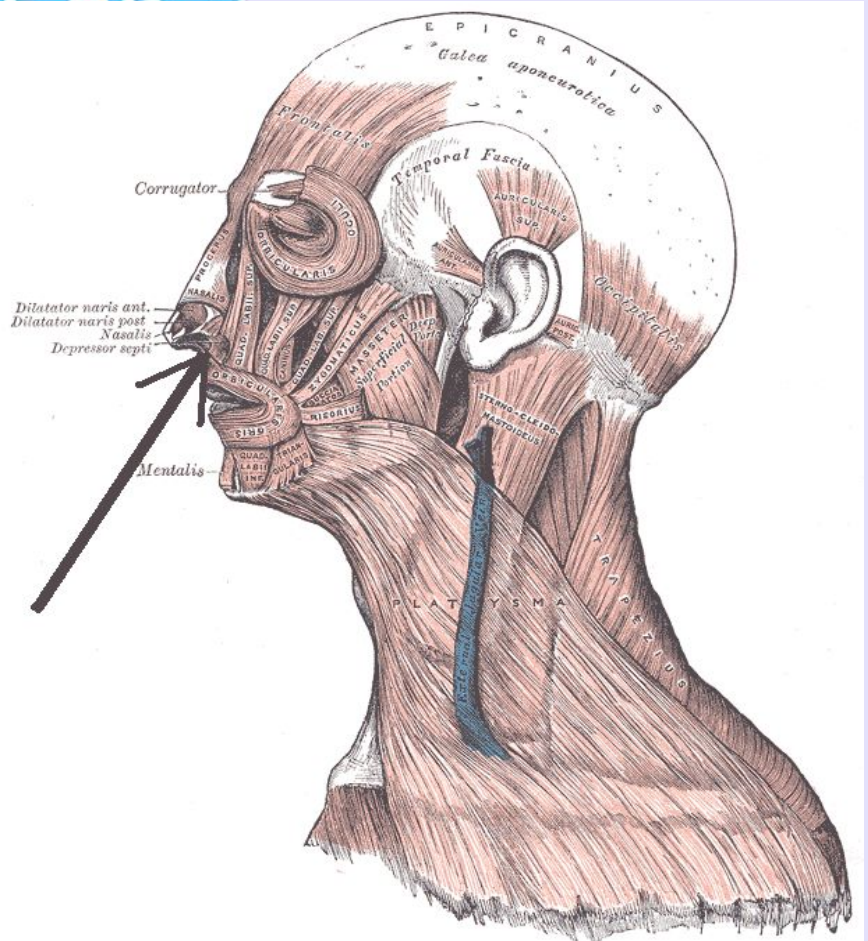
Nasalis

- **Nasalis.**
- It consists of two parts, transverse (compressor) and alar (dilator):
- **Compressor** (transverse part)
- Arises from the maxilla above and lateral to the incisive fossa; its fibers proceed upward and medially, expanding into a thin aponeurosis which is continuous on the bridge of the nose with that of the muscle of the opposite side, and with the aponeurosis of the Procerus.
 - compress the nasal cartilage.
- **Dilator** (alar part) is attached at one end to the greater alar cartilage, and at the other to the integument at the point of the nose.
 - Dilates the nasal cavity



Depressor septi

- Depressor septi is often regarded as part of dilator naris. It arises from the maxilla above the central incisor tooth and ascends to attach to the mobile part of the nasal septum.
- It is immediately deep to the mucous membrane of the upper lip.
- **Actions :**
Pulls the nasal septum downwards and, with the alar part of nasalis, widens the nasal aperture.



BUCCOLABIAL GROUP OF MUSCLES

- The shape of the mouth and the posture of the lips are controlled by a complex three-dimensional assembly of muscular slips.

- These include :

- **Elevators of the upper lip**

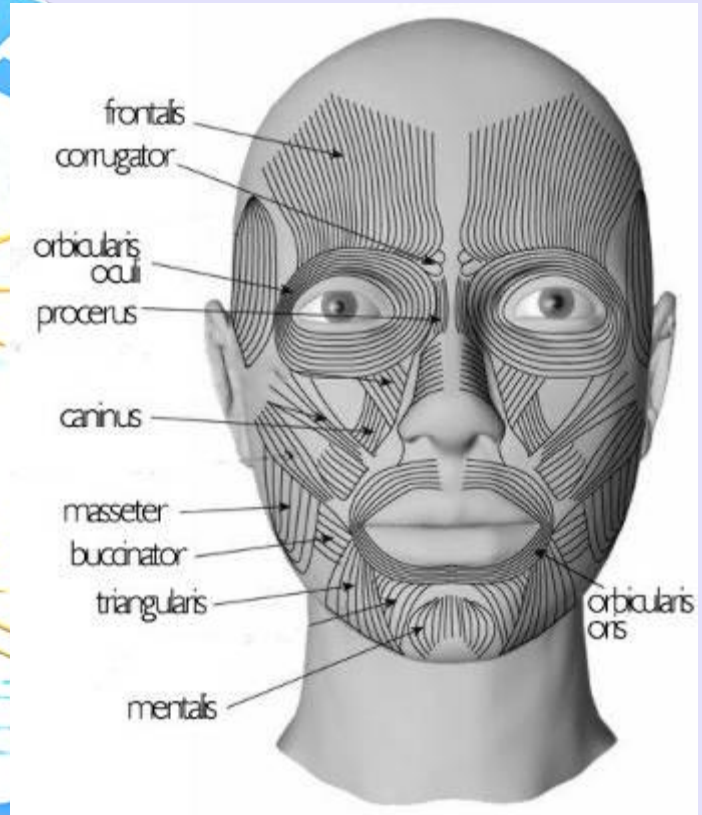
- Levator labii superioris alaeque nasi,
 - Levator labii superioris,
 - Zygomaticus major and minor,
 - levator anguli oris and
 - Risorius

- **Depressors, retractors and evertors of the lower lip .**

- Depressor labii inferioris
 - Depressor anguli oris
 - Mentalis

- **Sphincter muscles**

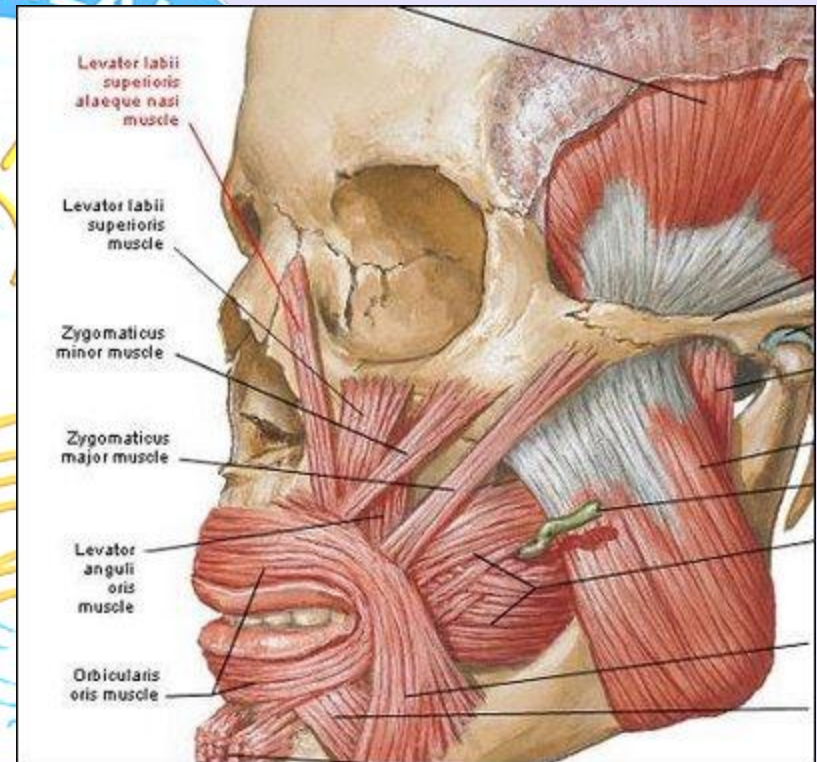
- Orbicularis oris,
 - Incisivus superior and inferior
 - Buccinator.



Levator labii superioris alaequae nasi

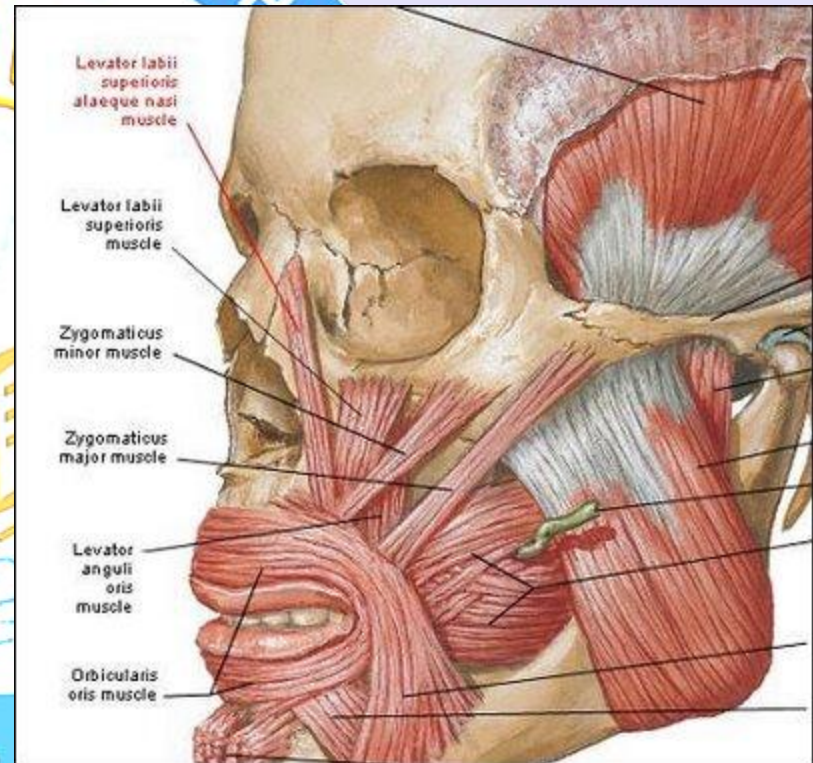


- Translated from Latin, the "lifter of the upper lip and of the wing of the nose". It has the longest name of any muscle.
- Attachment:
The muscle is attached to the upper frontal process of the maxilla and inserts into the skin of the lateral part of the nostril and upper lip.
- Actions:
 - It dilates the nostril and elevates the upper lip.



Levator labii superioris

- Levator labii superioris starts from the infraorbital margin.
- **Origin**
 - From the maxilla and zygomatic bone above the infraorbital foramen.
- **Insertion**
 - Its fibres converge into the muscular substance of the upper lip between the lateral slip of levator labii superioris alaequae nasi and zygomaticus minor.
- **Actions :**
 - Levator labii superioris elevates and everts the upper lip.



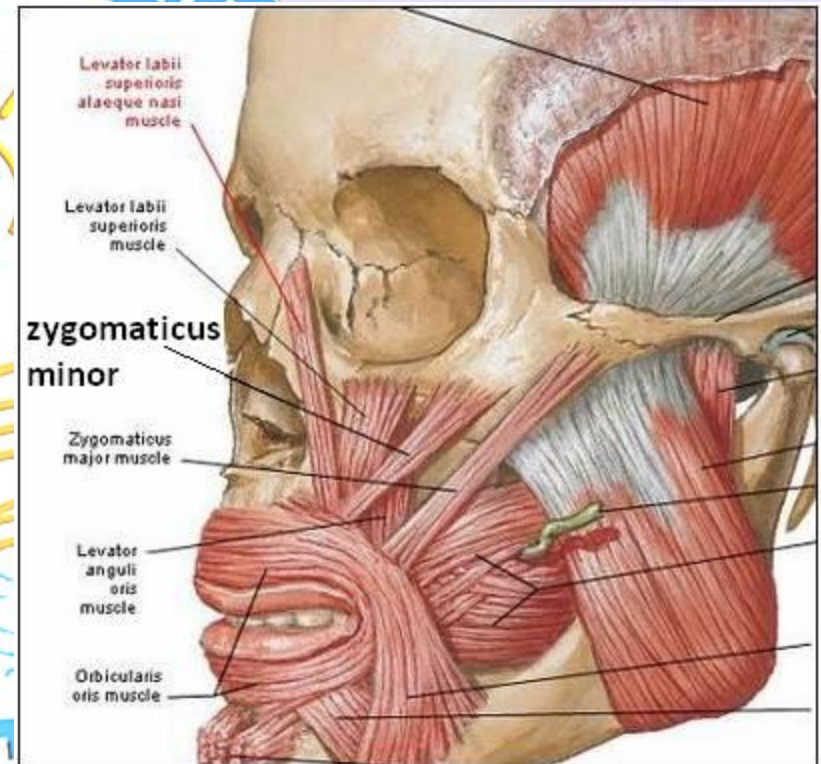
Zygomaticus major

- Zygomaticus major arises from the zygomatic bone. just in front of the zygomaticotemporal suture, and passes to the angle of the mouth.
- It blends with the fibres of levator anguli oris, orbicularis oris and more deeply placed muscular bands.
- Actions :
Zygomaticus major draws the angle of the mouth upwards and laterally as in laughing.



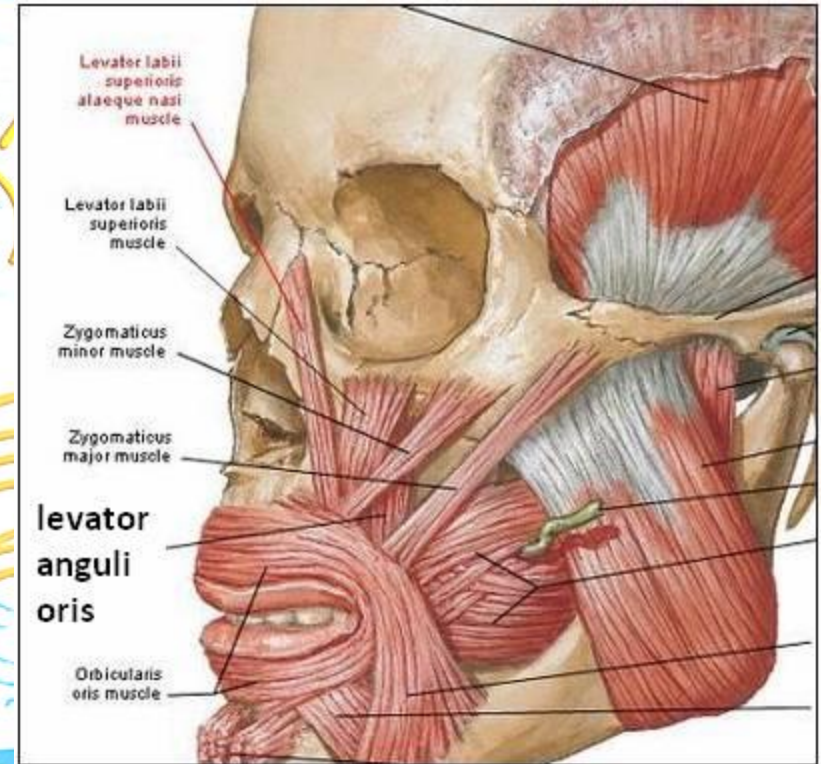
Zygomaticus minor

- Zygomaticus minor arises from the lateral surface of the zygomatic bone immediately behind the zygomaticomaxillary suture.
- passes downwards and medially into the muscular substance of the upper lip.
- **Actions :**
Elevates the upper lip, exposing the maxillary teeth.



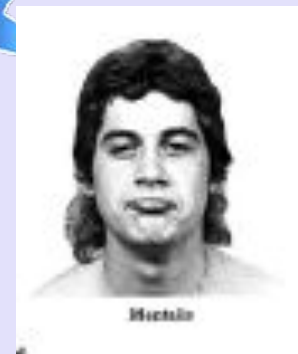
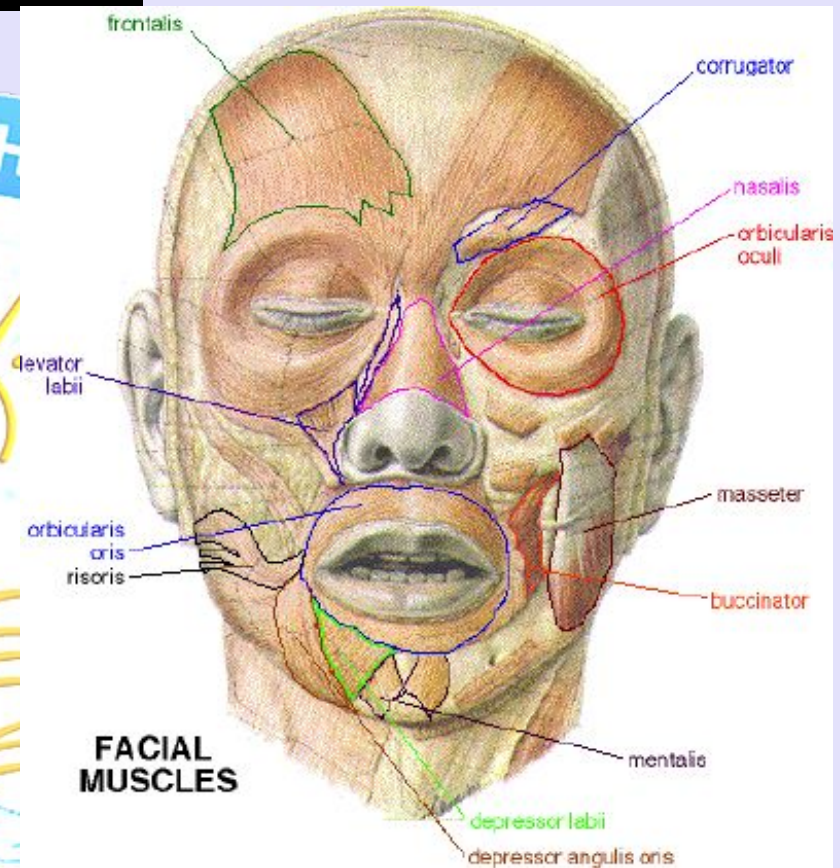
Levator anguli oris

- **levator anguli oris (caninus)** is a facial muscle of the mouth arising from the canine fossa, immediately below the infraorbital foramen.
- Its fibers are inserted into the angle of the mouth, intermingling with those of the Zygomaticus, Triangularis, and Orbicularis oris.
- **Actions :**
Raises the angle of the mouth in smiling, and contributes to the depth and contour of the nasolabial furrow.



Mentalis

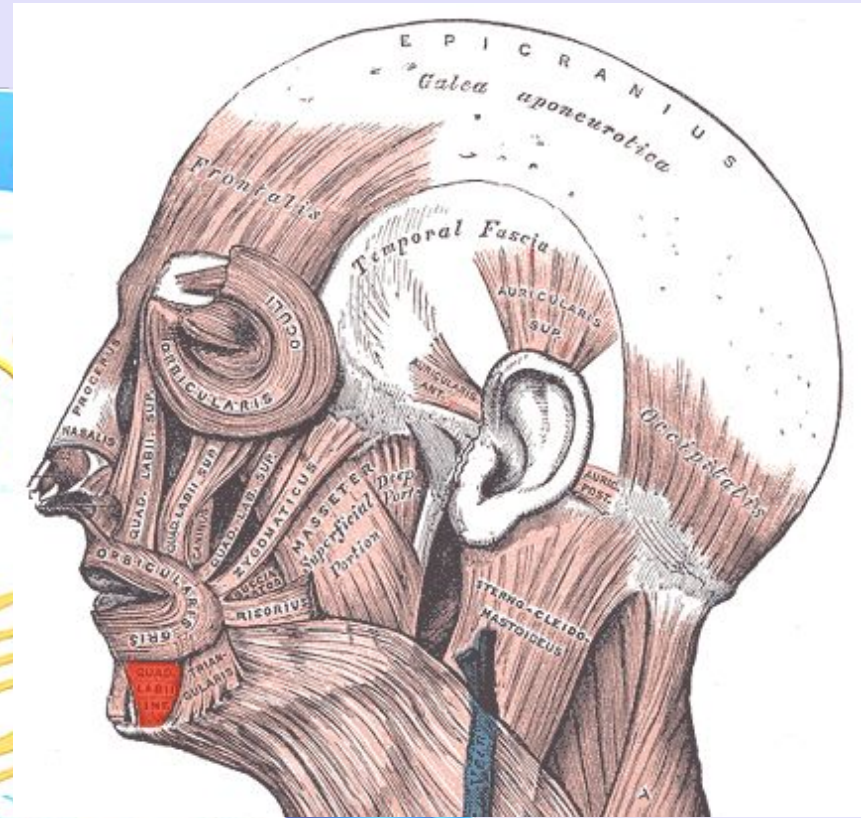
- Mentalis is a conical fasciculus lying at the side of the frenulum of the lower lip.
- The fibres arise from the incisive fossa of the mandible and descend to attach to the skin of the chin.
- **Actions :**
 - Mentalis raises the lower lip.
 - Wrinkling the skin of the chin.
 - It helps in protruding and everting the lower lip in drinking and also in expressing doubt or disdain.



Depressor labii inferioris

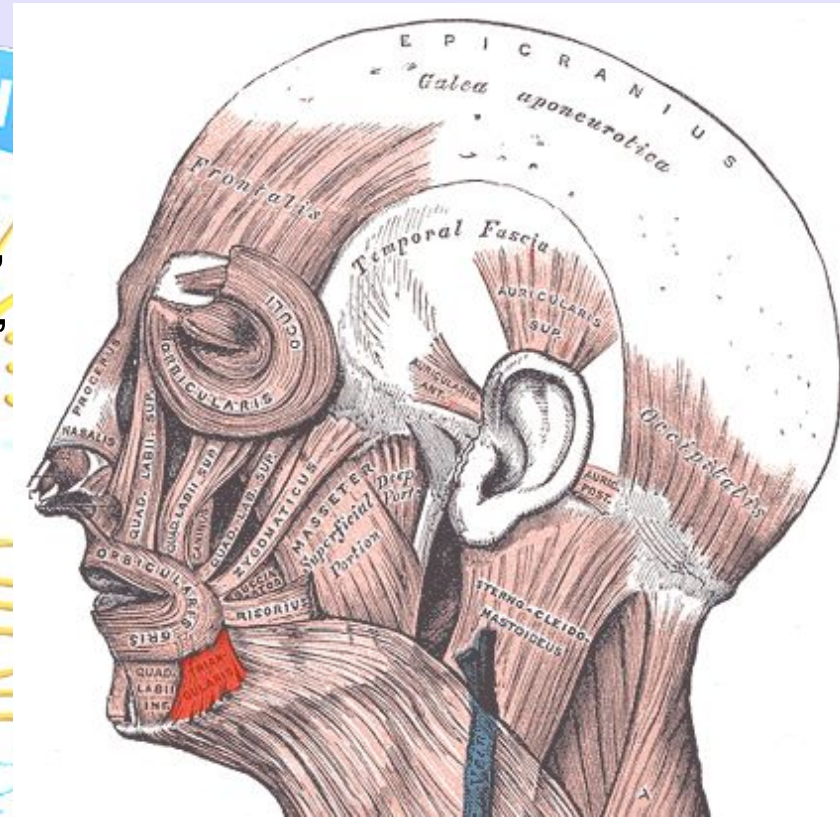


- Depressor labii inferioris is a quadrilateral muscle that arises from the oblique line of the mandible, between the symphysis menti and the mental foramen.
- It passes upwards and medially into the skin and mucosa of the lower lip.
- Blending with the paired muscle from the opposite side and with orbicularis oris.
- Below and laterally it is continuous with platysma.
- **Actions :**
Depress the lower lip.



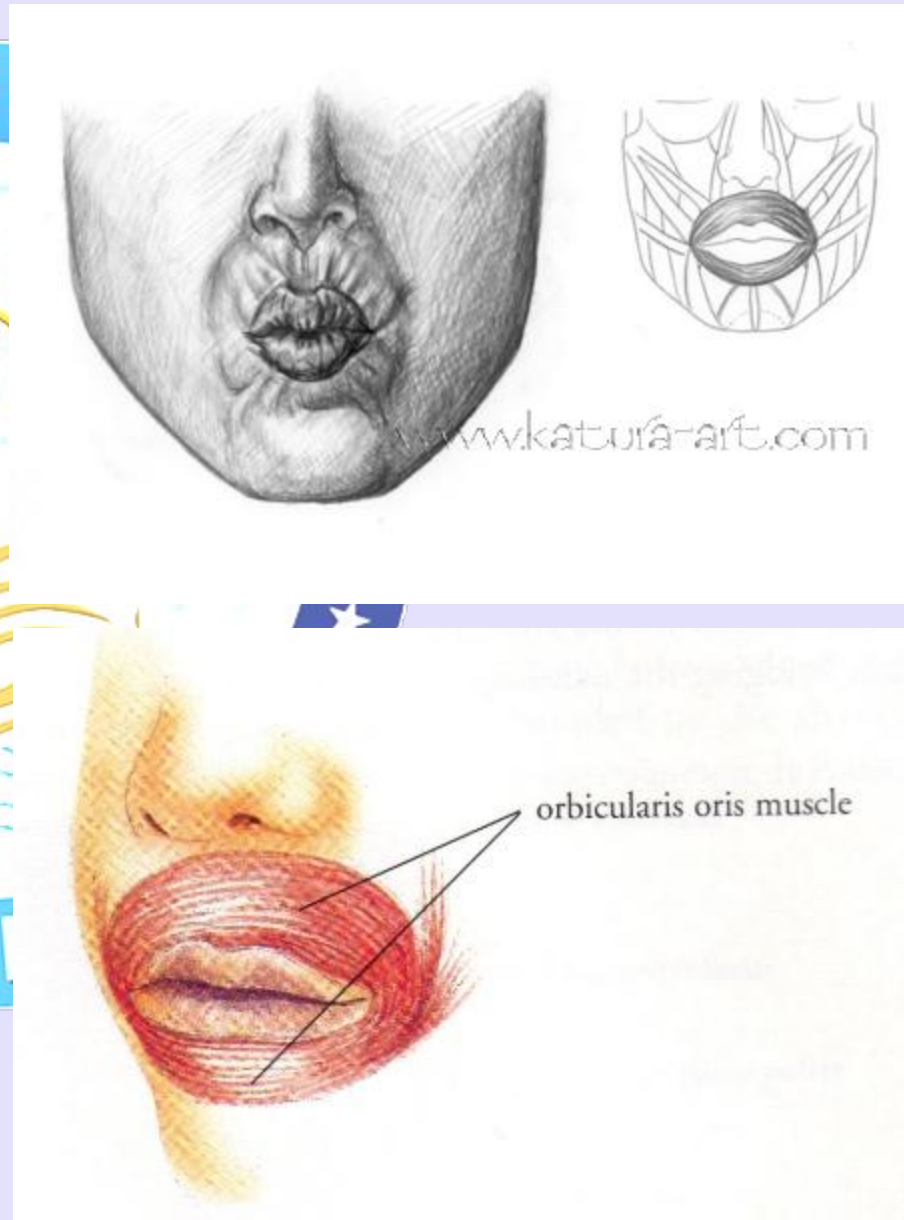
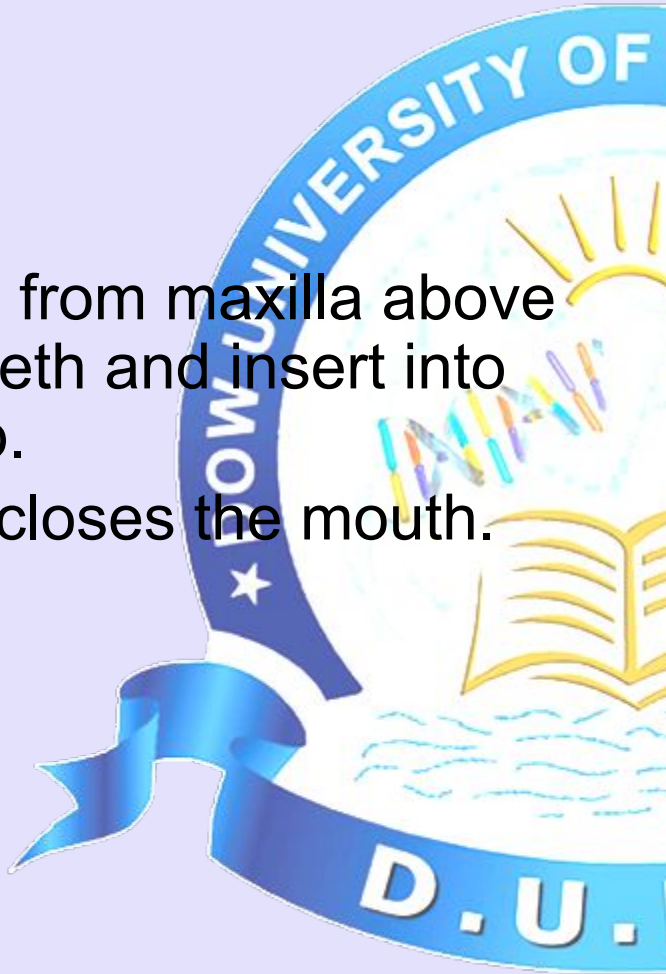
Depressor anguli oris

- Depressor anguli oris (Triangularis)
- **Origin and Insertion**
 - Arises from the oblique line of the mandible, whence its fibers converge, to be inserted, by a narrow fasciculus, into the angle of the mouth.
- At its origin it is continuous with the Platysma, and at its insertion with the Orbicularis oris and Risorius.
- **Actions :**
Angle of the mouth downwards and laterally in opening the mouth and in expressing sadness.



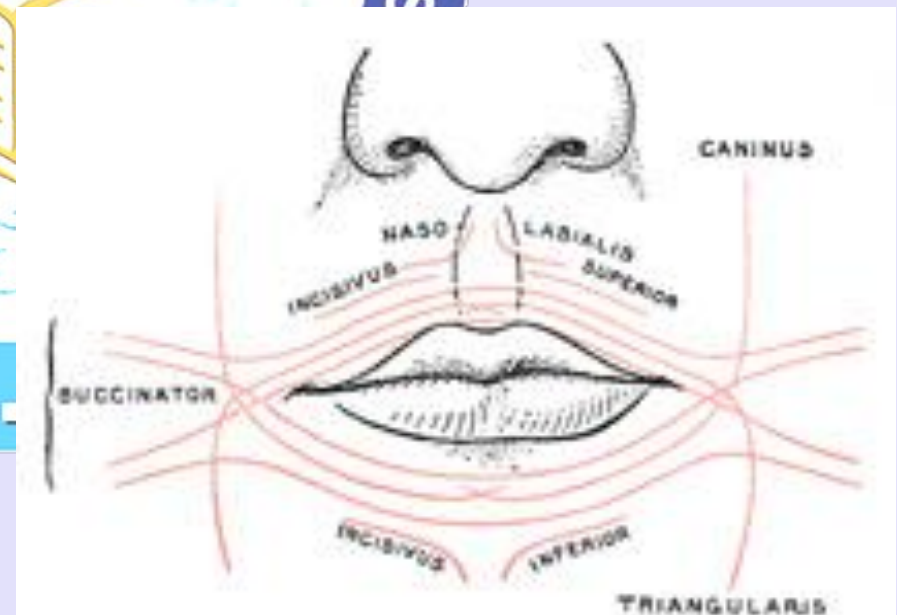
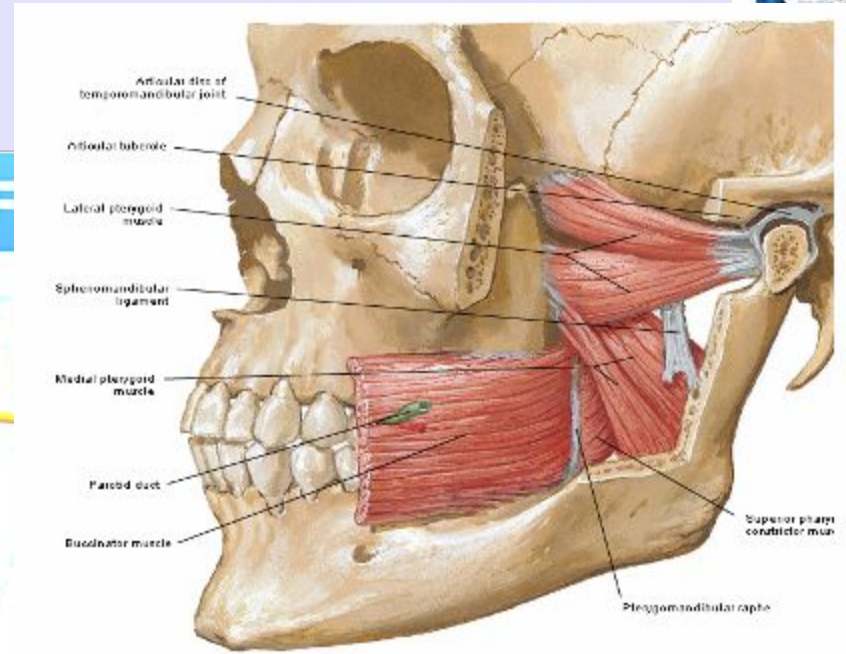
Orbicularis oris

- Originate from maxilla above incisor teeth and insert into skin of lip.
- **Action** –closes the mouth.



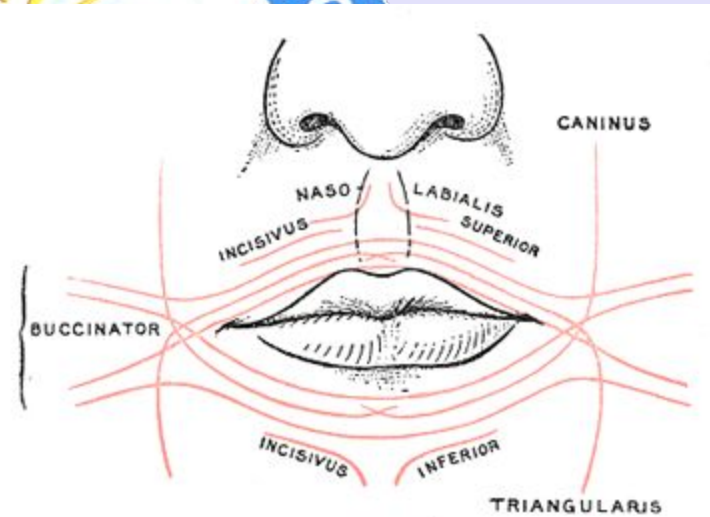
Buccinator

- Upper fiber:
 - Origin- from maxilla opposite molar teeth.
 - Insertion-upper lip.
- Lower fibers:
 - Origin-from mandible opposite molar teeth.
 - Insertion-lower lip.
- Middle fibers:
 - Origin –from pterigomandibular raphe.
 - Insertion-decussate before passing to lips.
- Action- prevent accumulation of food in vestibule of mouth.Flattens cheek against teeth



Incisivus labii superioris and inferioris

- Incisivus labii superioris has a bony origin from the floor of the incisive fossa of the maxilla above the eminence of the lateral incisor tooth.
- Its osseous attachment is to the floor of the incisive fossa of the mandible, lateral to mentalis and below the eminence of the lateral incisor tooth.



Platysma

- **Origin—**

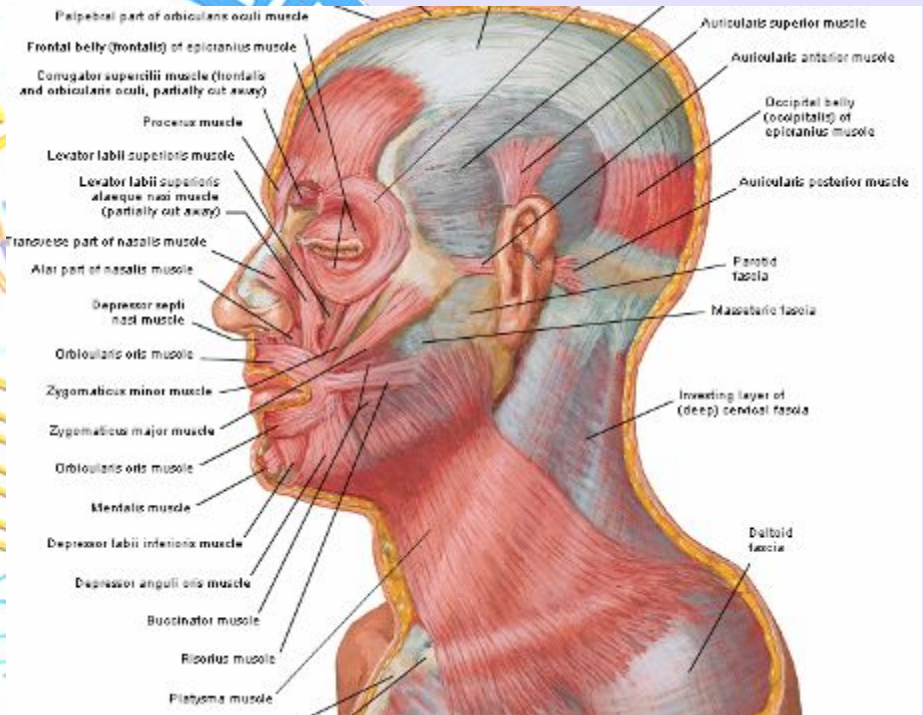
Upper part of pectoral and deltoid fascia.

- **Insertion—**

Base of mandible, skin of lower face and lip.

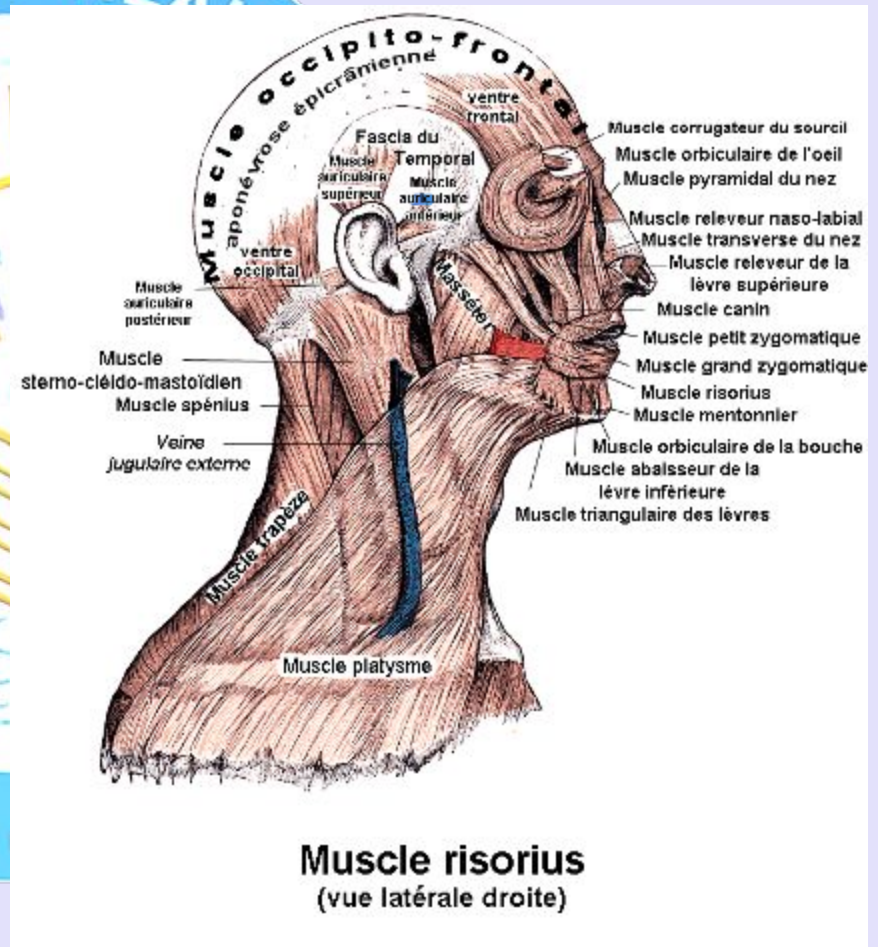
- **Action—**

Releases pressure of skin on the subjacent veins, depress mandible, pulls angle of mouth downwards.



Risorius

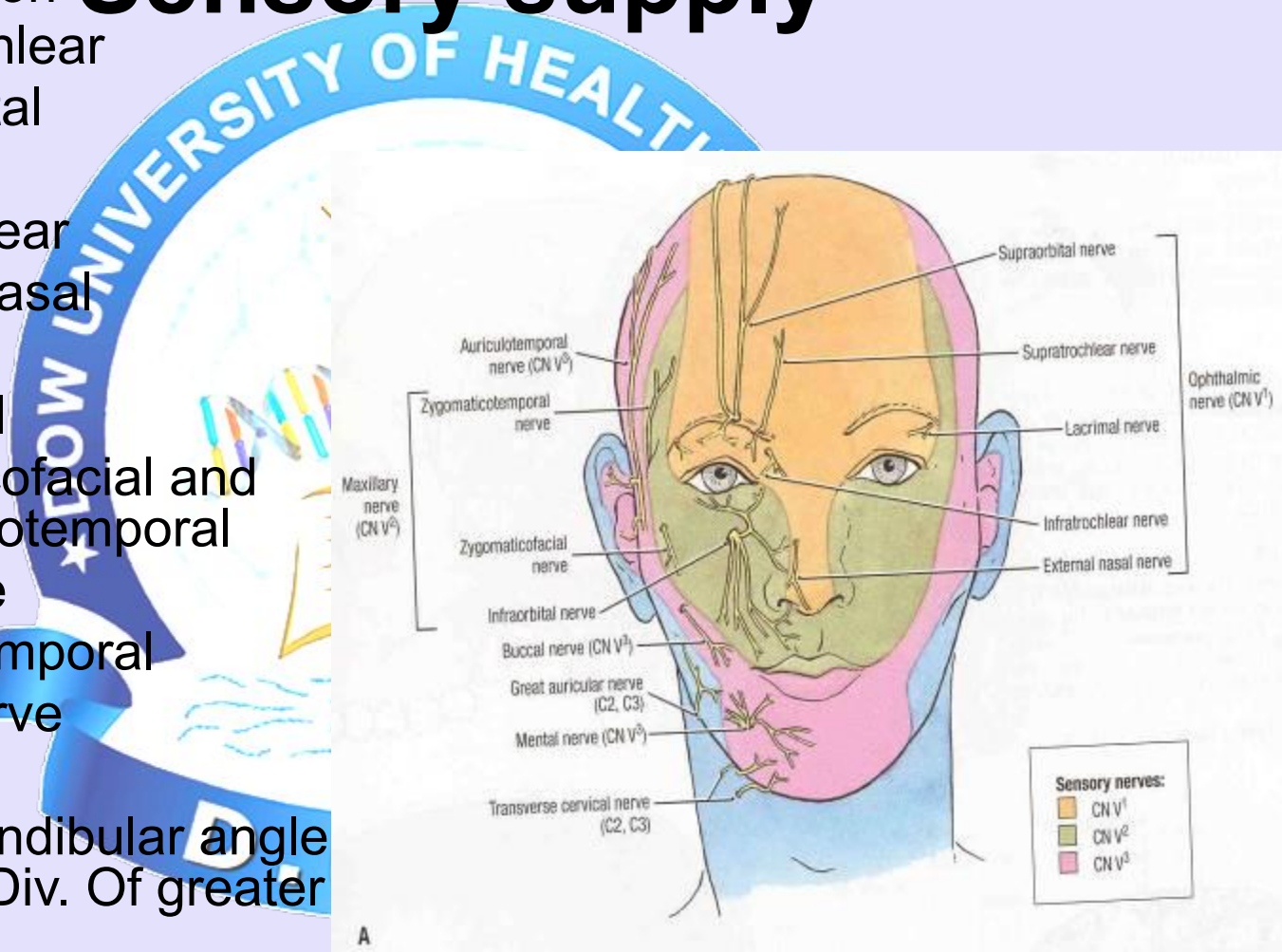
- **Origin**
 - Arises in the fascia over the parotid gland.
- **Insertion**
 - Passing horizontally forward, superficial to the platysma.
 - inserts onto the skin at the angle of the mouth.
- **Actions :**
Pulls the corner of the mouth laterally in numerous facial activities, including grinning and laughing



Nerve supply of face

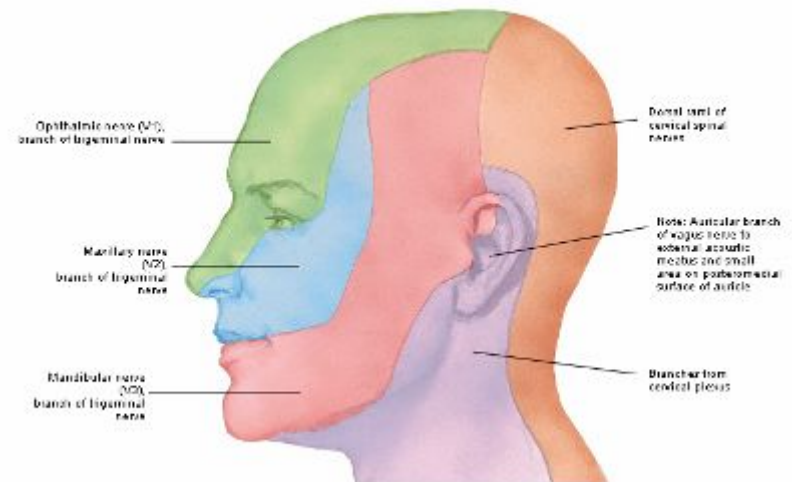
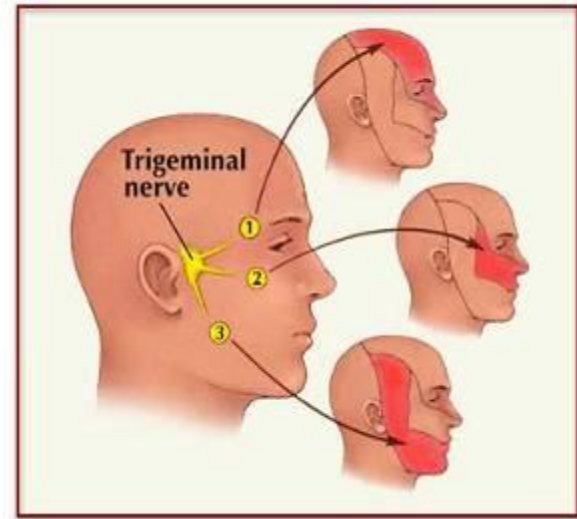
Sensory supply

- Ophthalmic division
 - Supratrochlear
 - Supraorbital
 - Lacrimal
 - Infratrochlear
 - External nasal
- Maxillary nerve
 - Infraorbital
 - Zygomaticofacial and zygomaticotemporal
- Mandibular nerve
 - Auriculotemporal
 - Buccal nerve
 - Mental
- Skin over the mandibular angle supplied by ant. Div. Of greater auricular n.



Trigeminal nerve

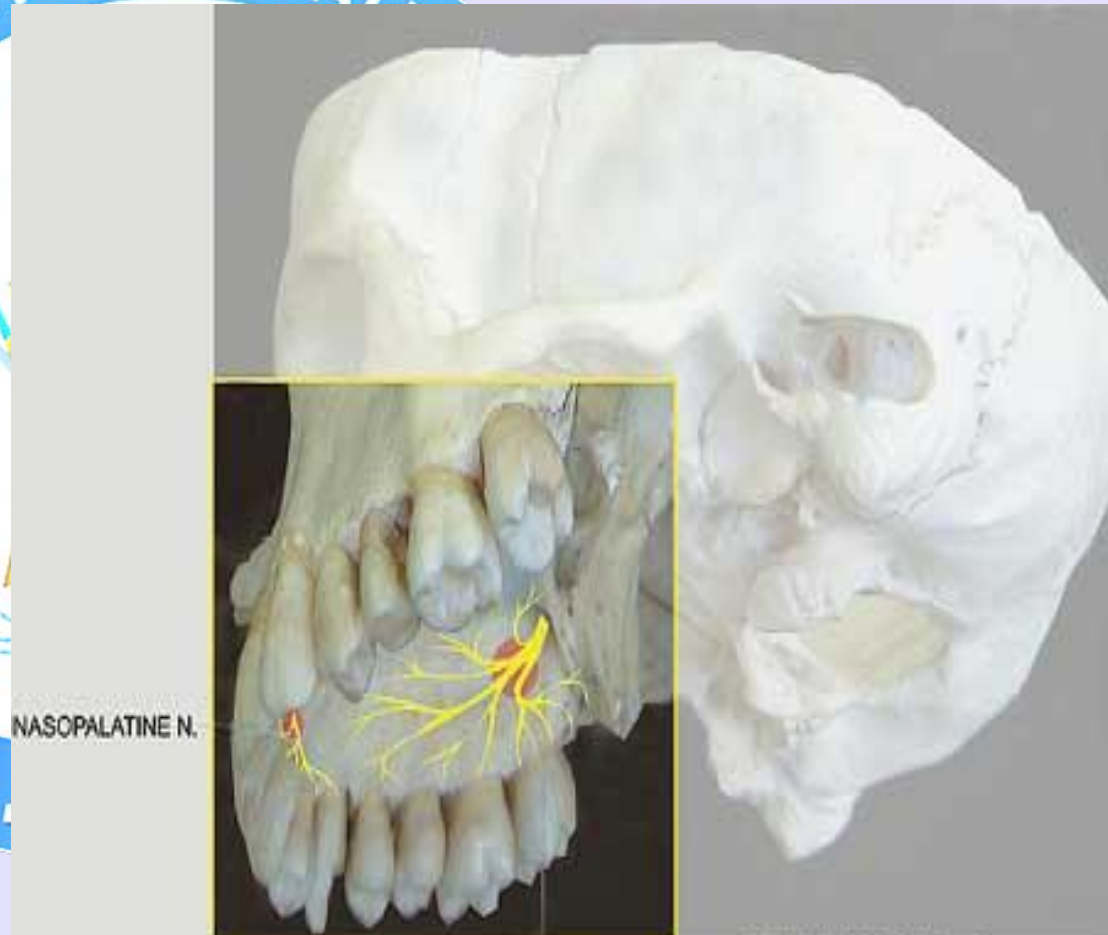
- The trigeminal nerve is the primary source of innervations for the oral cavity.
- The trigeminal nerve subdivided into three main branches:
 - Ophthalmic.
 - Maxillary.
 - Mandibular.



Trigeminal nerve- Maxillary division

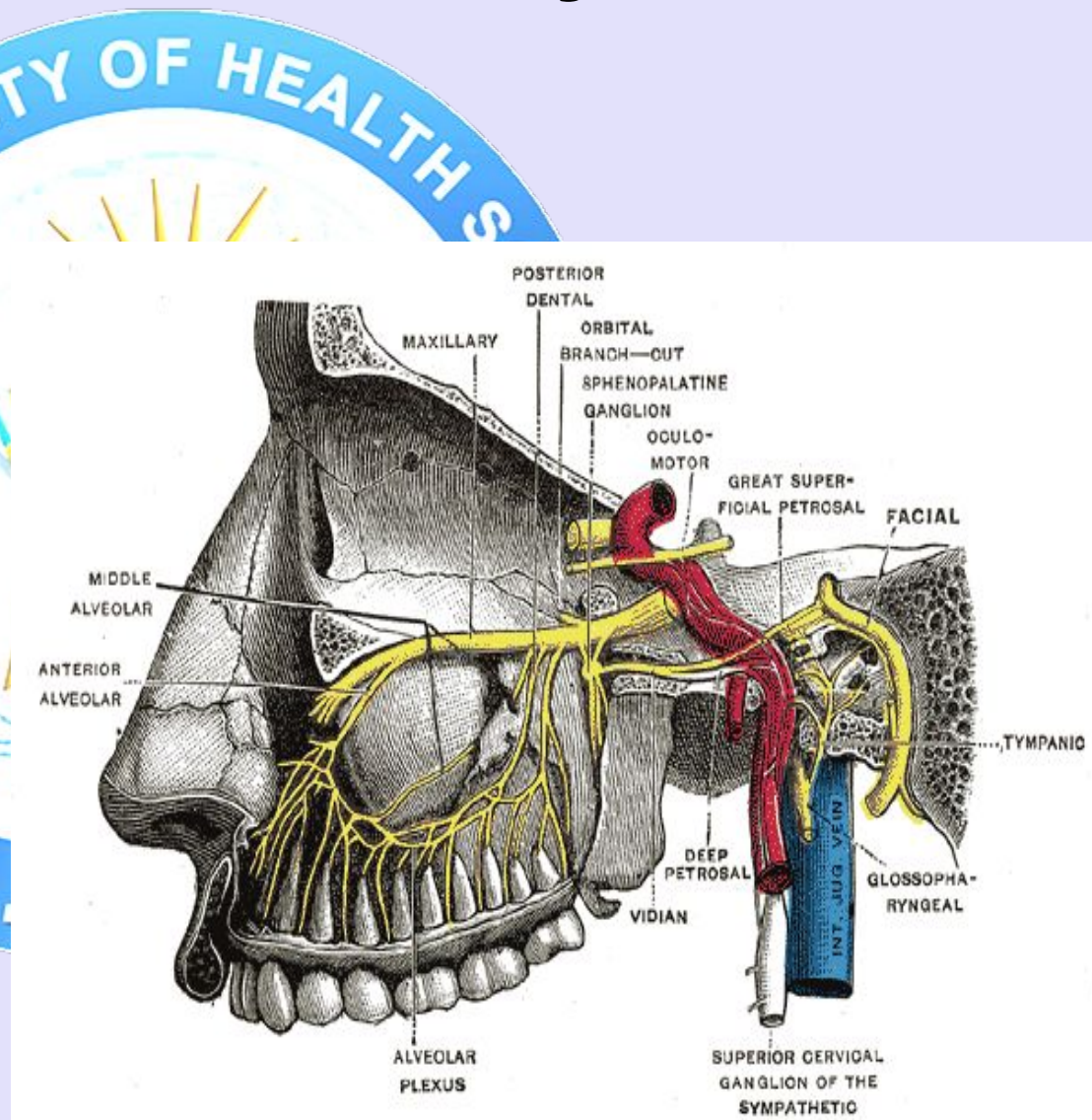
- The maxillary division subdivides to provide the following innervations:

- **1: The nasopalatine** nerve, which passes through the incisive foramen, supplying the mucoperiosteum (periosteum covered with mucous membrane) palatal to the maxillary anterior teeth.



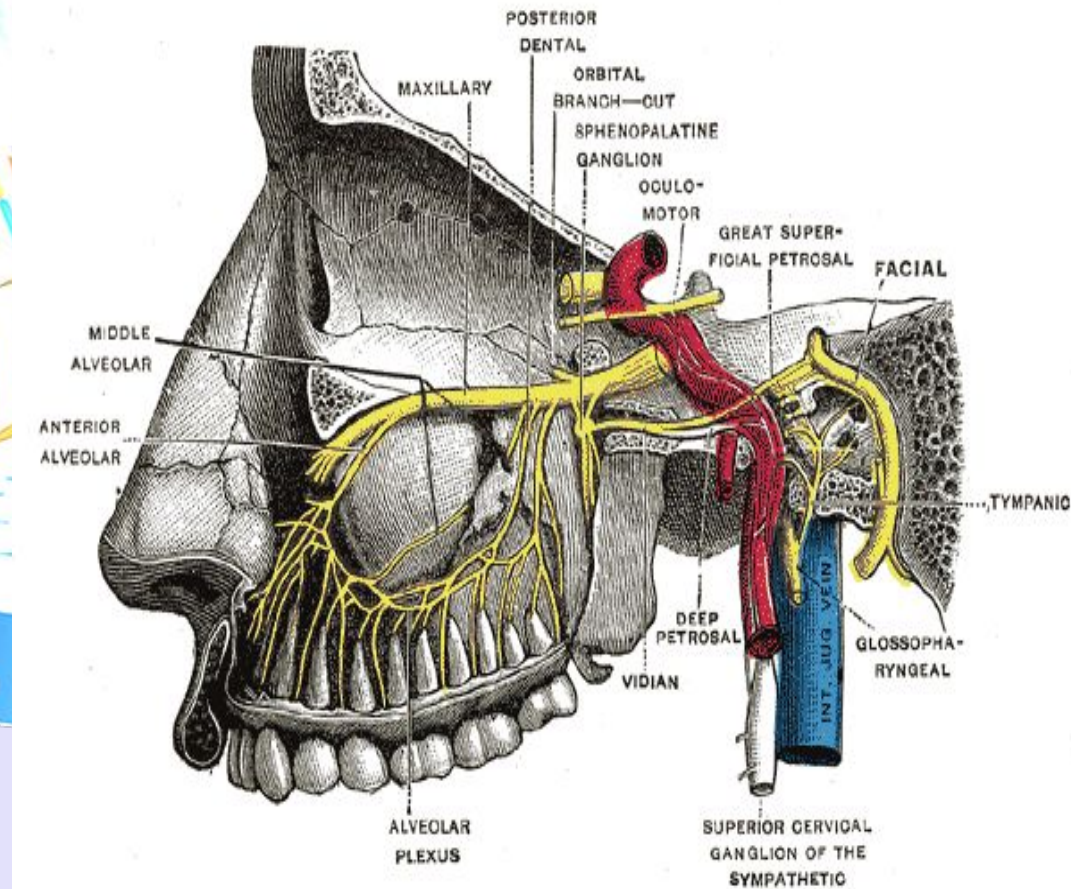
Trigeminal nerve- Maxillary division

- **2. The anterior palatine** nerve, which passes through the posterior palatine foramen and forward over the palate, supplying the mucoperiosteum intermingling with nasopalatine nerve.
- **3. The anterior superior alveolar** nerve supplies the maxillary central lateral and cuspid teeth, plus their periodontal membrane and gingiva. This nerve also supplies the maxillary sinus.



Trigeminal nerve- Maxillary division

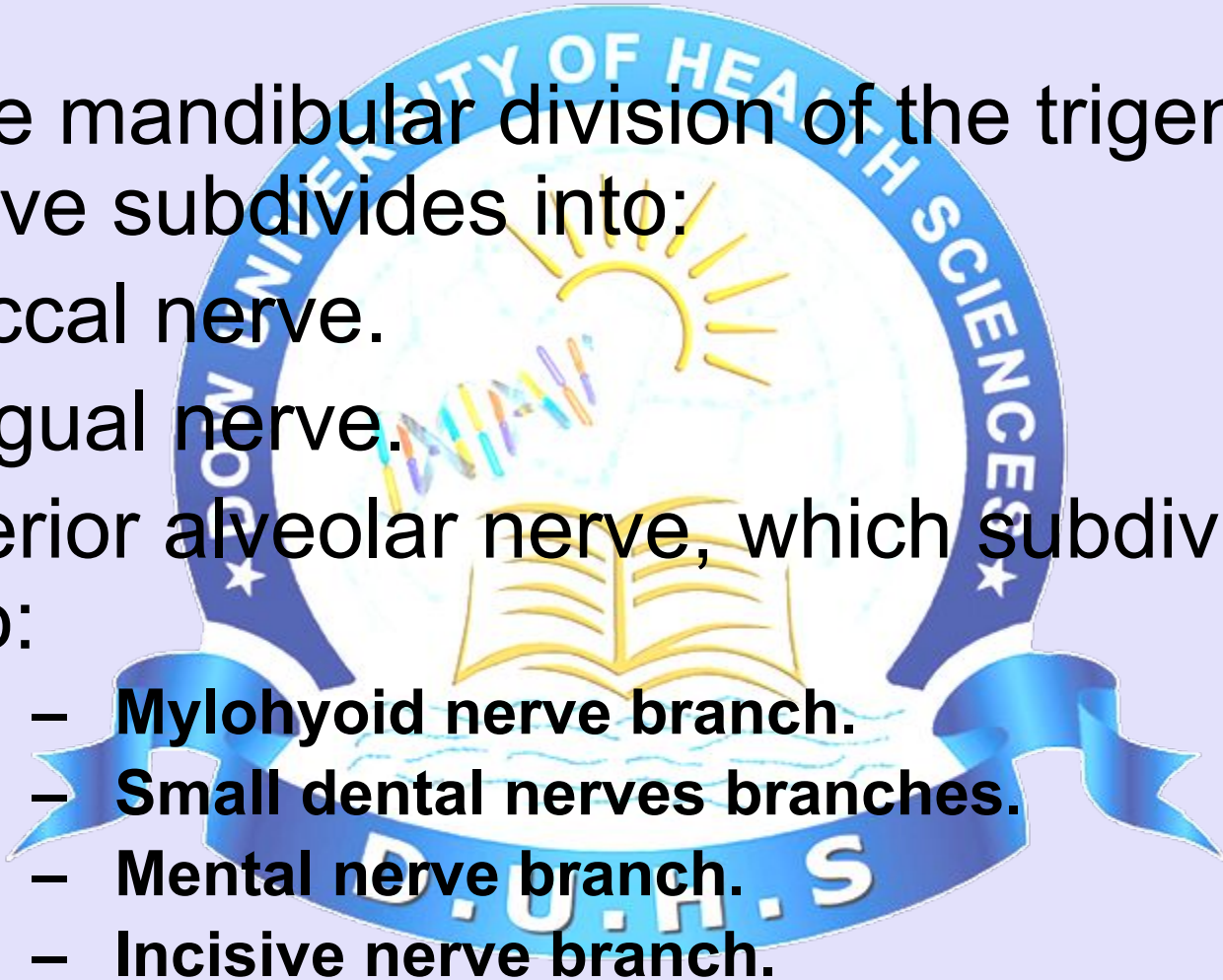
- **4. The middle superior alveolar** nerve supplies the maxillary first and second premolars, the mesiobuccal root of the maxillary first molar and maxillary sinus.
- **5. The posterior superior alveolar** nerve supplies the other roots of the maxillary first molar and the maxillary second and third molars, it also branches forward to serve the lateral wall of the maxillary sinus.



Trigeminal nerve- Mandibular division

The mandibular division of the trigeminal nerve subdivides into:

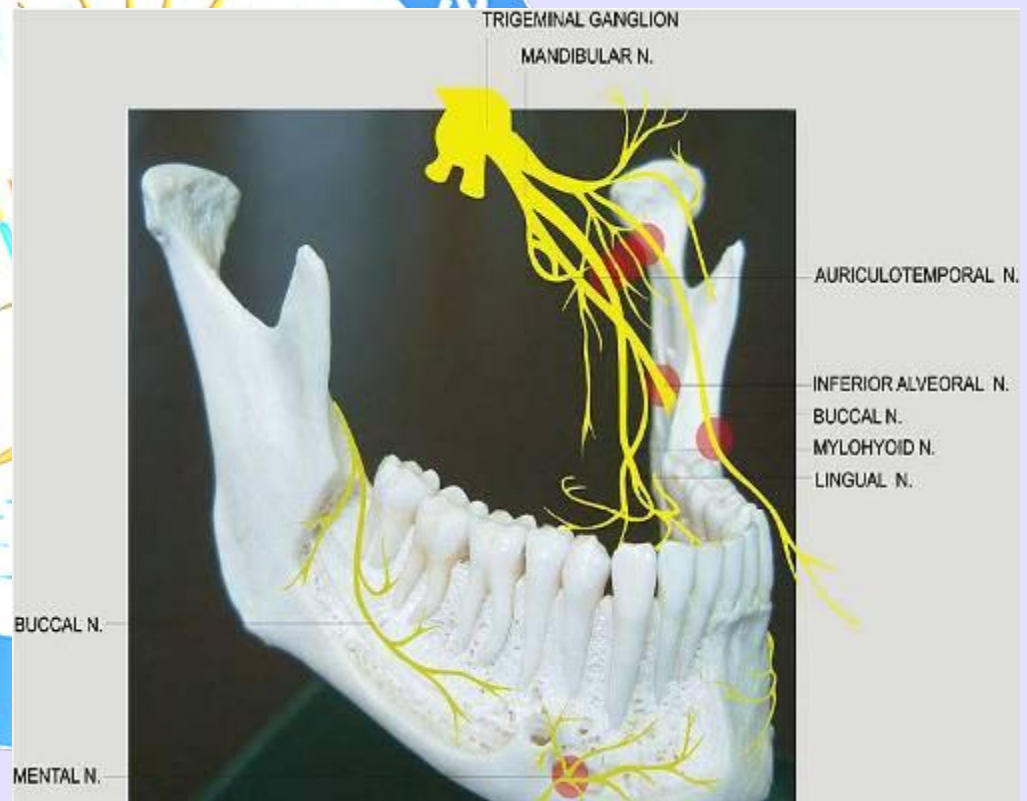
- 1) Buccal nerve.
- 2) Lingual nerve.
- 3) Inferior alveolar nerve, which subdivides into:
 - Mylohyoid nerve branch.
 - Small dental nerves branches.
 - Mental nerve branch.
 - Incisive nerve branch.



Trigeminal nerve - Mandibular division

1. The buccal nerve supplies branches to the buccal mucous membrane and the mucoperiosteum of the maxillary and mandibular molar teeth.

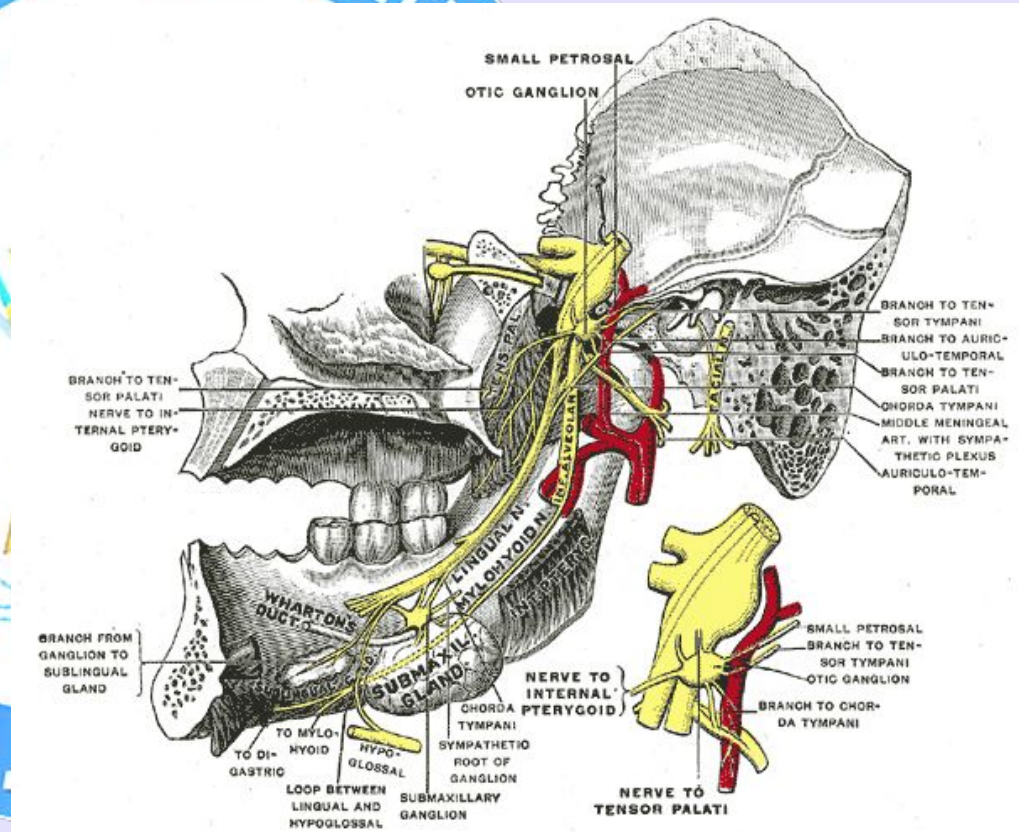
2. The lingual nerve supplies the anterior 2/3 rd of the tongue and gives off branches to supply the lingual mucous membrane and mucoperiosteum.



Trigeminal nerve - Mandibular division

3. The inferior alveolar nerve subdivides into

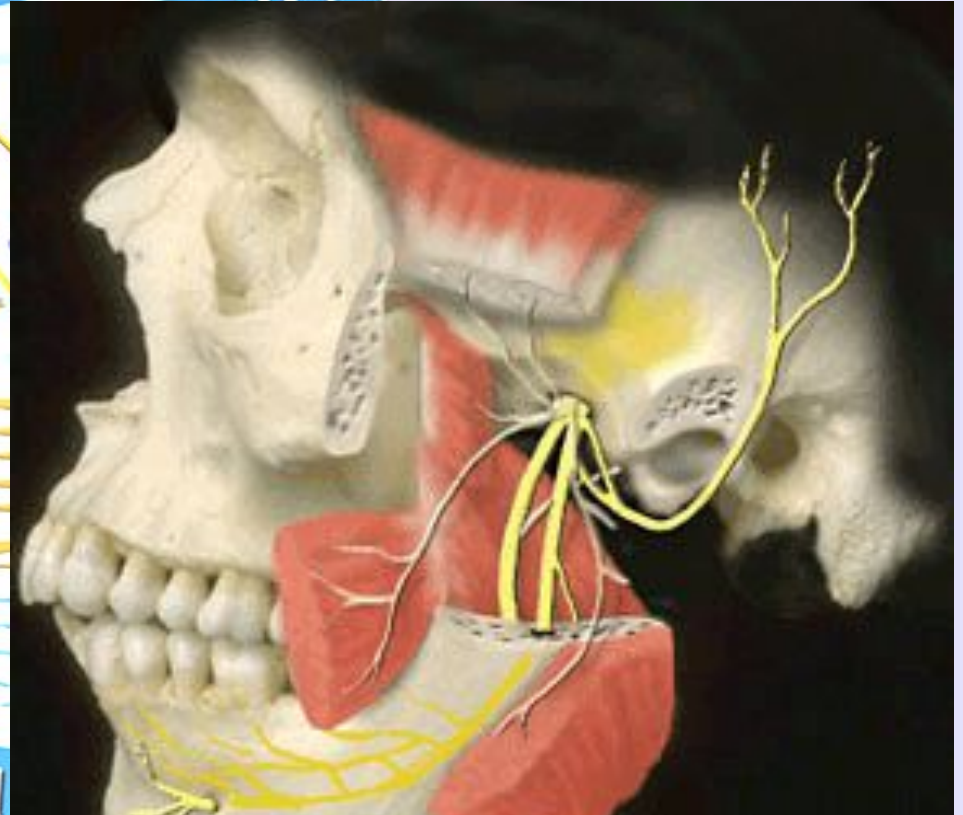
- Mylohyoid nerve supplies the mylohyoid muscles and the anterior belly of the digastric's muscle.
- Small dental nerves supply the molar and premolar teeth, alveolar process and periosteum.



Trigeminal nerve - Mandibular division



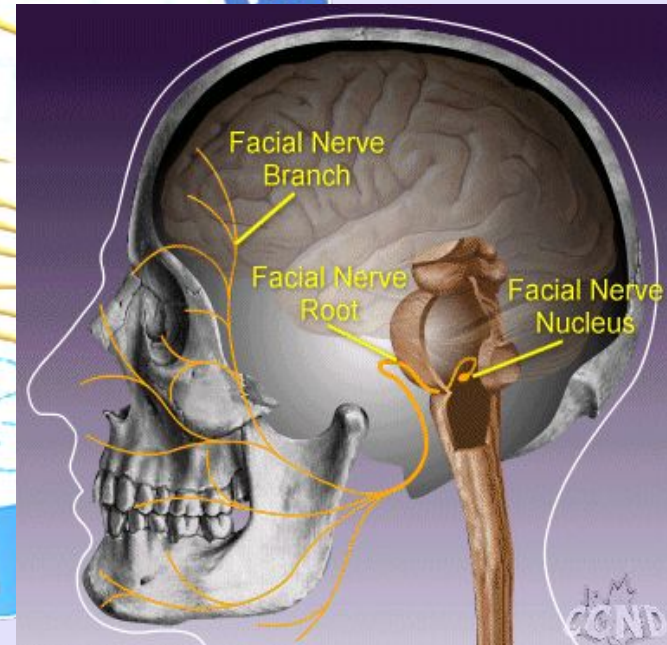
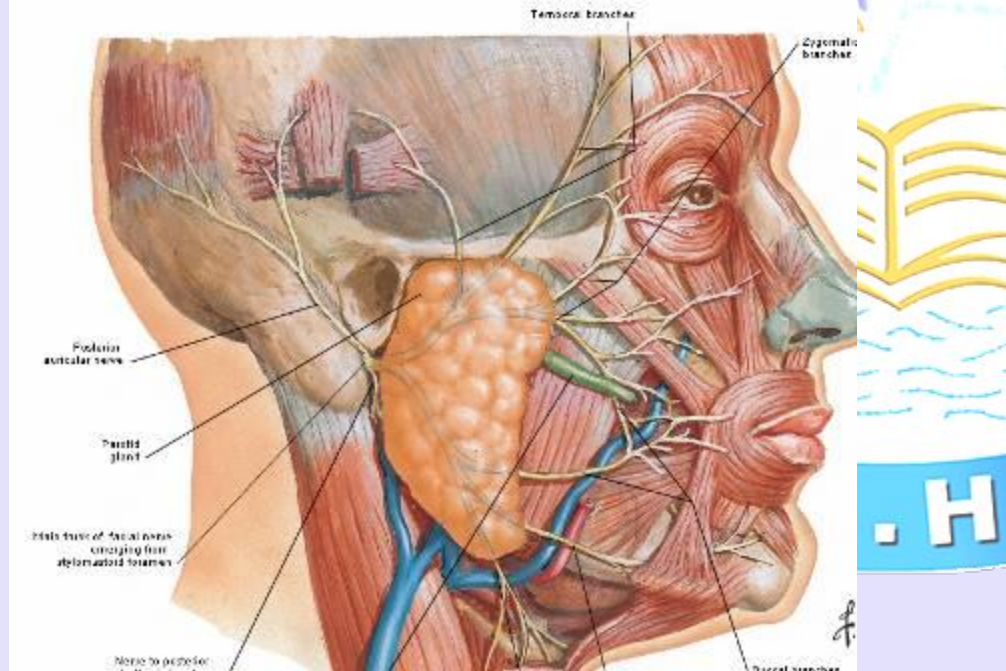
- Mental nerve, which moves outward and anteriorly through the mental foramen and supplies the chin and mucous membrane of the lower lip.
- Incisive nerve, which continues anteriorly within the bone and gives off small branches to supply the cuspid, lateral and central incisors.



Nerve supply of face

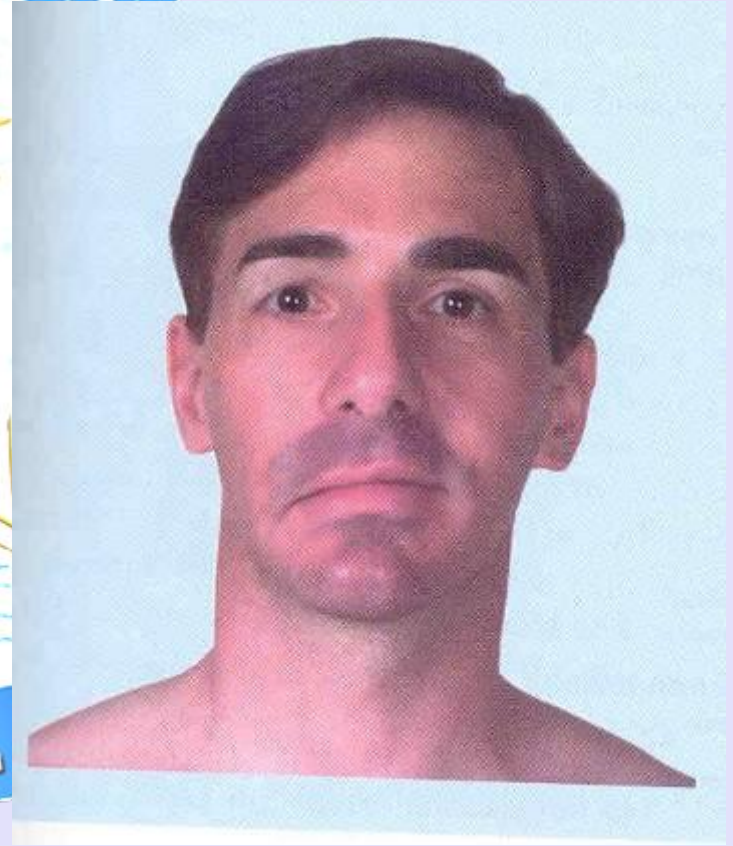
- Motor supply:
 - Facial nerve innervate the muscles of facial expression, salivary glands, lacrimal glands & sense of taste on the anterior 2/3 rd of the tongue.

Facial Nerve Branches and Parotid Gland in Situ



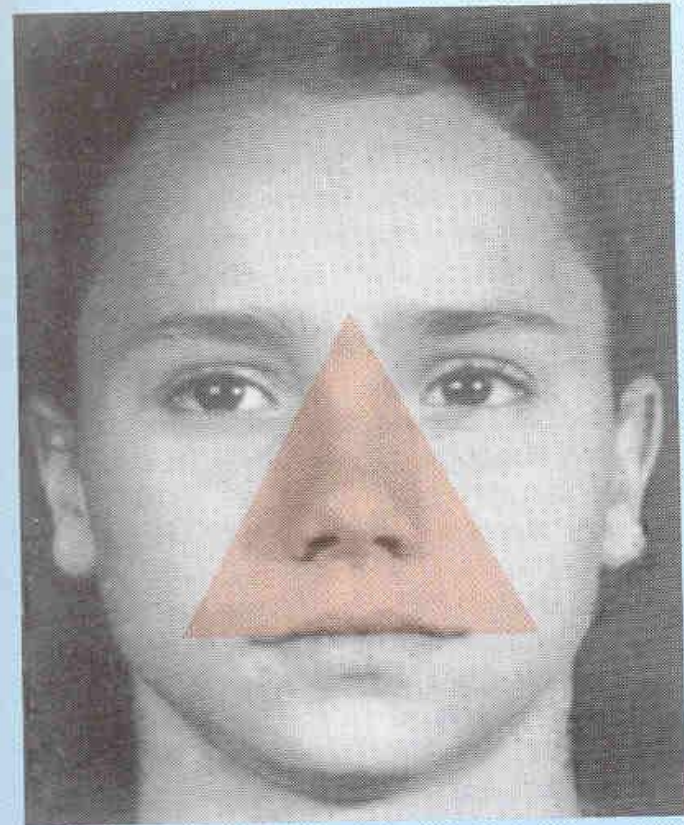
Applied

- **Trigeminal neuralgia:**
 - Maxillary and mandibular nerve are involved.
 - Excruciating pain in the region of distribution of these nerve.
- In infranuclear lesions of facial nerve (eg, **Bell's palsy**) :- whole face is paralyzed
 - c/f
 - Affected side is motionless
 - Loss of wrinkles
 - Eye cannot be closed
 - In smiling the mouth is drawn to normal side
 - During mastication food accumulates in vestibule of mouth
- In supranuclear lesions of facial nerve only the lower part of face is paralyzed. The upper part (frontalis & part of orbicularis oculi) escapes due to its bilateral innervation.



- Dangerous area of face

Infections from face mainly from upper lip & nose can go to cavernous sinus through ophthalmic vein and deep facial vein.



(B)

